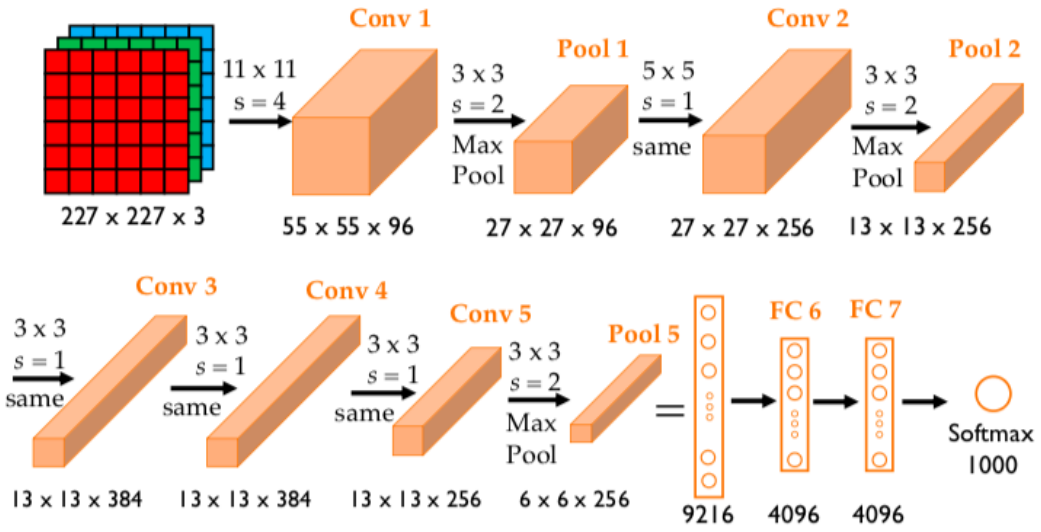
Generalized Image Classification

Recall: A convolution neural network for image classification:



These are often trained on *ImageNet*, which means they are built to classify into ≈ 1000 categories . . . How to generalize this to other classification problems?

Idea: Each layer learns *features* that appear in images. The goal is to use this information to to simplify training for a new classification task. How?

Transfer Learning

Transfer learning is a machine learning technique in which knowledge gained through one task or dataset is used to improve model performance on another related task or different dataset.

Some approaches for Images:

- ConvNet as fixed feature extractor
- Fine-tuning the ConvNet
- Pretrained models

When to use what?

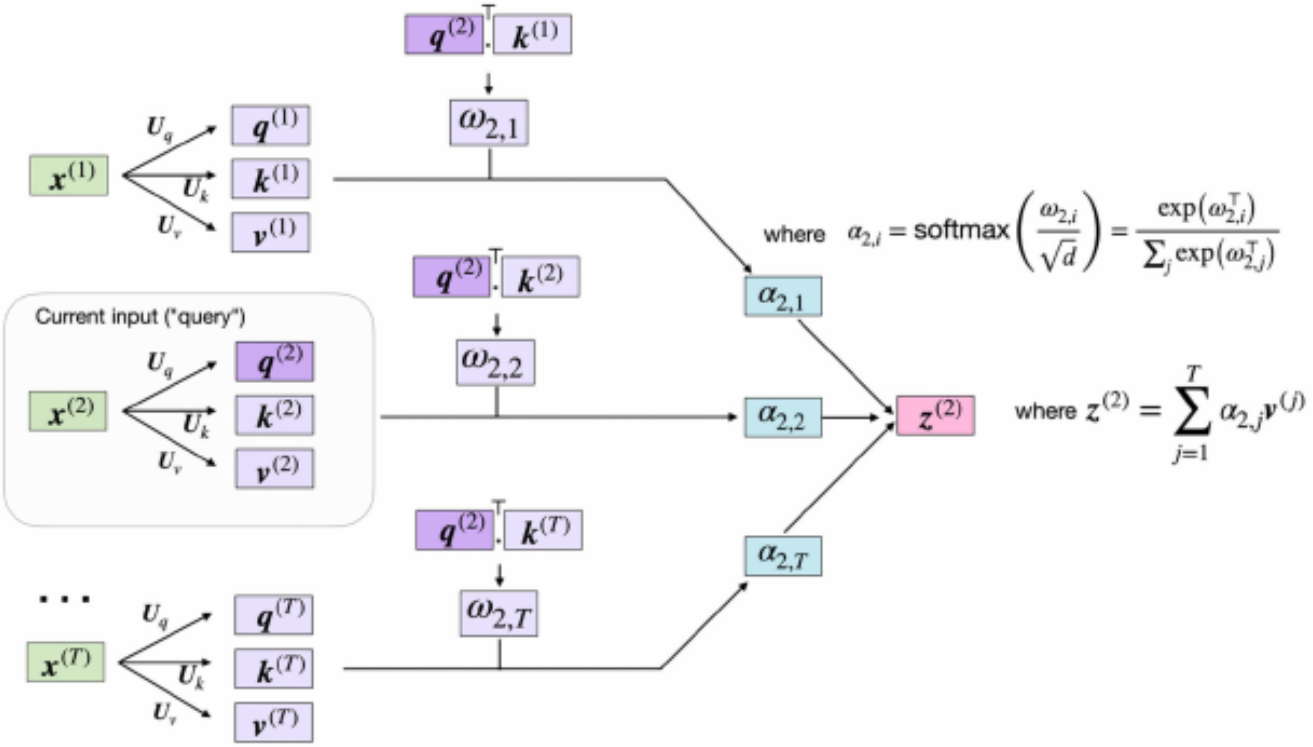
- New dataset is small, similar to original $\implies$ train just a linear classifier (last layer)
- New dataset is large and similar to the original dataset $\implies$ fine-tune
- New dataset is small, very different from the original $\implies$ linear classifier

See https://cs231n.github.io/transfer-learning/ for more.

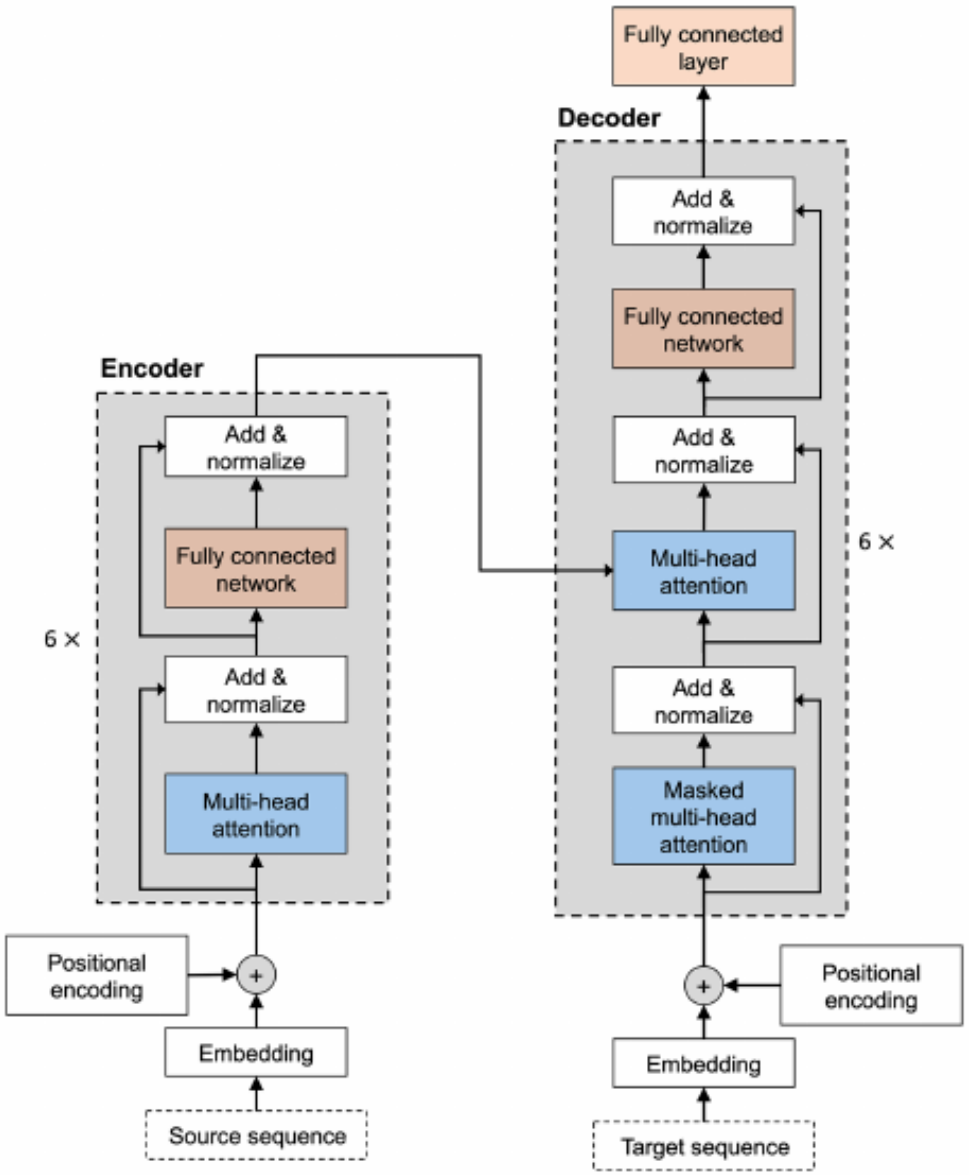
Transfer Learning: Pytorch Images

```
model_ft = models.resnet18(weights='IMAGENET1K_V1')
num_ftrs = model_ft.fc.in_features
# Here the size of each output sample is set to 2.
# Alternatively, it can be generalized to ``nn.Linear(num_ftrs, len(class_names))``.
model_ft.fc = nn.Linear(num_ftrs, 2)

model_ft = model_ft.to(device)

criterion = nn.CrossEntropyLoss()

# Observe that all parameters are being optimized
optimizer_ft = optim.SGD(model_ft.parameters(), lr=0.001, momentum=0.9)

# Decay LR by a factor of 0.1 every 7 epochs
exp_lr_scheduler = lr_scheduler.StepLR(optimizer_ft, step_size=7, gamma=0.1)
```

Recall Transformer



Recall Transformer

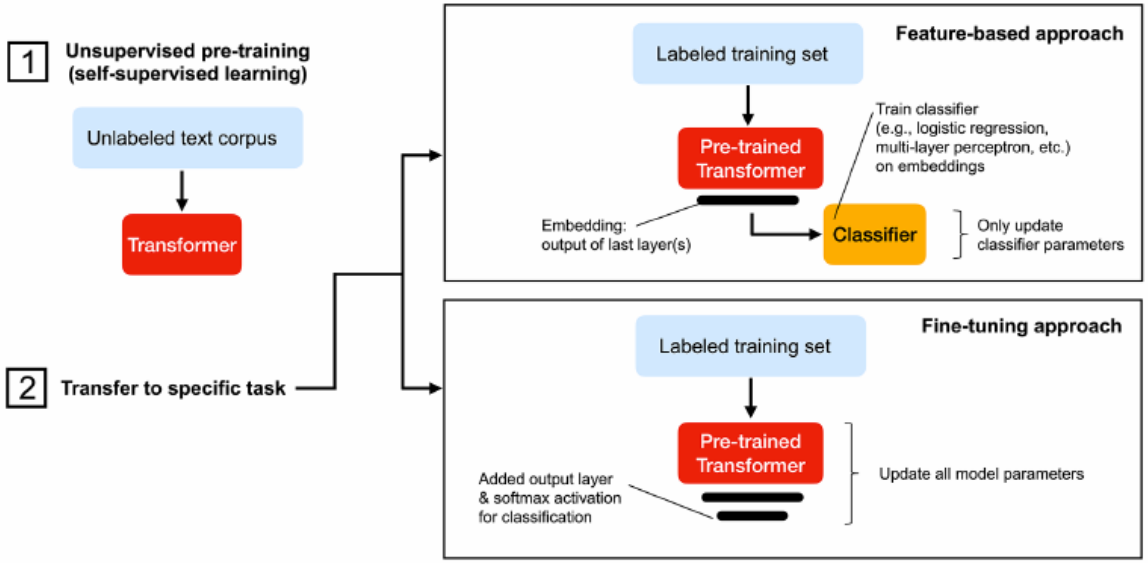


Transformer Success

Idea: Most of the training can be done with *unlabelled* data using **self-supervised learning** to perform **next-word** prediction.

1. At time *step 1*, feed in the ground-truth words 1, ..., i-1.
2. Ask the model to predict the word at position i and compare it with the ground-truth word i.
3. Update the model and time step, $i := i+1$. Go back to step 1 and repeat until all words are processed.

Transfer Learning?

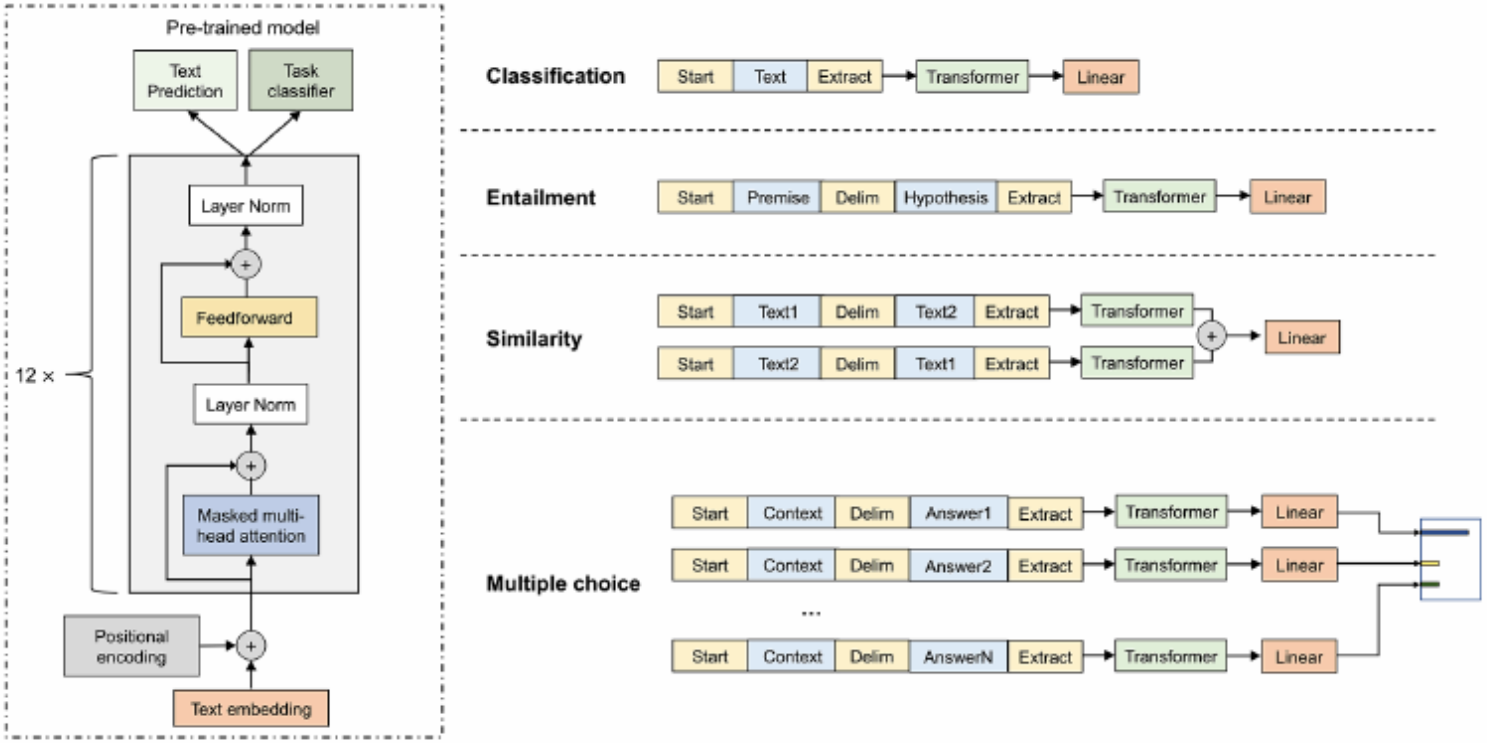


The GPT Architecture

Training GPT (1.1):

1. Pre-train on a large amount of unlabeled plain text
2. Supervised fine-tuning

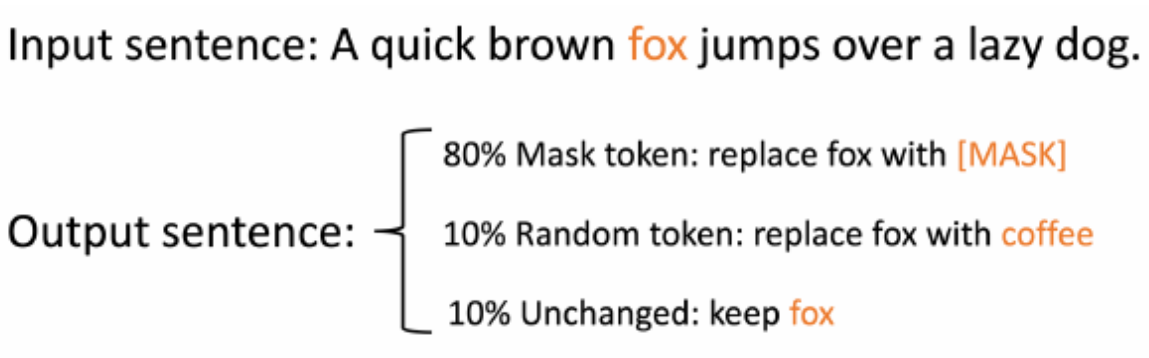
Types of Fine-Tuning:



The BERT Architecture

BERT is **Bidirectional Encoder Representations from Transformers**.

Idea: BERT adopts a masked-language model, replacing random tokens by [MASK]. 15 percent of words are marked for masking . . .



Idea: Bidirectional training disables BERT's ability to generate a sentence word by word but provides input encodings of higher quality for other tasks, such as classification, since the model can now process information in both directions.

There's also **BART:** BART can be viewed as a generalization of both GPT and BERT. As the title of this section suggests, BART is able to accomplish both tasks, generating and classifying text.